DB Types

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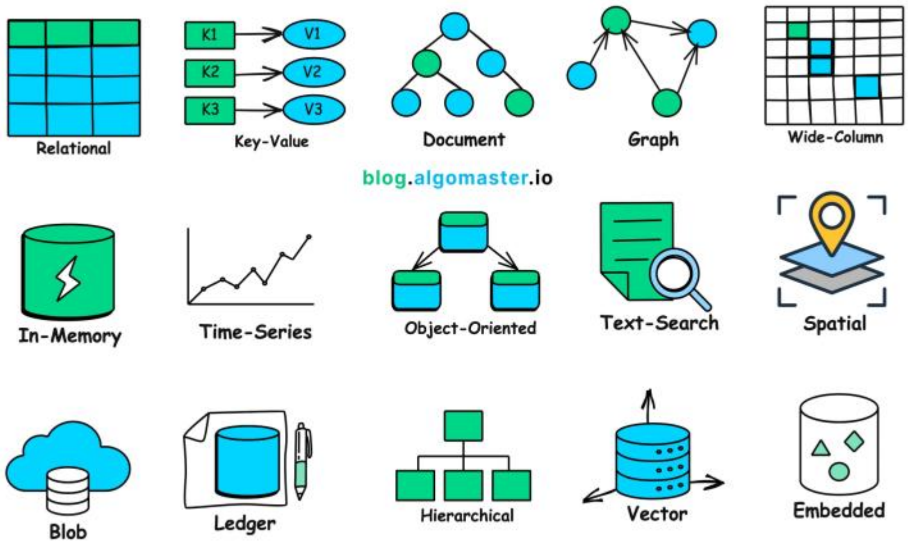

15 Types of Databases

1. Relational Database (RDBMS)

Idea: Data is stored in **tables (rows + columns)**.

- Tables can be connected using **foreign keys**.
- Supports **SQL**.
- Strong data integrity.
- Supports **ACID transactions** and complex queries.

Use when:

- Data is structured.
- Relationships between data are important.
- Transactions must be reliable.

Examples: MySQL, PostgreSQL, Oracle

Remember:

👉 **RDBMS = Tables + Relationships + SQL + ACID**

2. Key-Value Store

Idea: Store data as:

key → value

Example:

user:101 → "John"

- Very fast lookup using a unique key.
- Highly scalable.
- Good for simple data access.

Use cases:

- Sessions
- Caching
- Real-time data

Examples: Redis, DynamoDB

Remember:

👉 **Key-Value = Fast lookup by key**

3. Document Database

Idea: Store data as documents, usually **JSON/BSON**.

Example:
```
{
  "name": "John",
  "age": 25,
  "skills": ["Java", "SQL"]
}
```
- Flexible schema.
- Easy to change the structure of data.
- Good read/write performance.

Use when:
- Data structure changes frequently.
- Data naturally looks like JSON/documents.

Examples: MongoDB, Couchbase
Remember:
👉 **Document DB = JSON-like flexible data**

4. Graph Database
Idea: Store data as:
Nodes + Edges + Properties
Example:
John → FRIEND_OF → Mike
Mike → WORKS_AT → Google
Great for finding relationships between entities.
Use cases:
- Social networks
- Recommendation systems
- Knowledge graphs

Examples: Neo4j, Amazon Neptune
Remember:
👉 **Graph DB = Relationships are the main focus**

5. Wide-Column Store
Idea: Designed for **huge distributed datasets** across many machines.
- Flexible columns.
- High scalability.
- High write throughput.
- Often uses eventual consistency.

Use cases:
- Web analytics
- User activity tracking
- Real-time analytics

Examples: Cassandra, HBase, Bigtable
Remember:
👉 **Wide-Column = Huge distributed data + high writes**

6. In-Memory Database
Idea: Data is stored primarily in **RAM instead of disk**.
RAM is much faster than disk, so access is extremely fast.
Use cases:
- Caching
- Gaming
- Real-time systems
- High-frequency trading

Examples: Redis, Memcached
Remember:
👉 **In-Memory = RAM = Very Fast**

7. Time-Series Database
Idea: Designed for data that changes over **time**.
Example:
10:00 → CPU = 40%
10:01 → CPU = 45%
10:02 → CPU = 60%
Use cases:
- Monitoring
- Stock prices
- IoT sensors
- Performance metrics

Examples: InfluxDB, TimescaleDB, Prometheus
Remember:
☞ **Time-Series = Data + Timestamp**

8. Object-Oriented Database
Idea: Stores data as **objects**, similar to objects in Java/C++/Python.
Example:
```
User Object
├── name
├── age
└── methods
```
Useful when application data is naturally represented as objects.
Examples: ObjectDB, db4o
Remember:
☞ **OODB = Database stores objects**

9. Text Search Database
Idea: Optimized for searching large amounts of **text**.
Instead of simple exact matching, it can efficiently search and index text.
Use cases:
- Product search
- Web search
- Log analysis

Examples: Elasticsearch, Solr, Sphinx
Remember:
☞ **Text Search DB = Search through lots of text**

10. Spatial Database
Idea: Designed for **location/geographical data**.
Can work with:
Points
Lines
Polygons
Example:
User location → (17.3850, 78.4867)
Use cases:
- Maps
- Finding nearby places
- Route optimization
- Vehicle tracking

Examples: PostGIS, Oracle Spatial
Remember:
☞ **Spatial DB = Location + Geography**

11. Blob Datastore

Idea: Stores **large unstructured files**.
Examples:
Images
Videos
Audio
Documents
Backups
Designed for massive amounts of data with high durability and scalability.
Use cases:

- Image/video storage
- CDN content
- Backups
- Archives

Examples: Amazon S3, Azure Blob Storage, HDFS
Remember:
👉 **Blob = Big files**

12. Ledger Database

Idea: Maintains an **immutable, append-only history**.
Once a record is stored, it should not be changed or deleted.
Example:
Transaction 1
Transaction 2
Transaction 3
...
Useful when you need a trustworthy history of changes.
Use cases:

- Supply chains
- Healthcare records
- Voting systems

Examples: Amazon QLDB, Hyperledger Fabric
Remember:
👉 **Ledger = Cannot secretly change history**

13. Hierarchical Database

Idea: Data is organized like a **tree**.
Company
├── Engineering
│ ├── Backend
│ └── Frontend
└── HR
Each record has one parent but can have multiple children.
Use cases:

- Organizational structures
- File systems

Examples: IBM IMS, Windows Registry
Remember:
👉 **Hierarchical = Tree structure**

14. Vector Database

Idea: Stores **vectors (arrays of numbers)** and searches for similar vectors.
Example:
"cat" → [0.21, 0.54, 0.81, ...]
"kitten" → [0.23, 0.51, 0.79, ...]
If vectors are close, the underlying data is considered similar.
Use cases:

- AI/ML

- Semantic search
- Image search
- Recommendations
- Anomaly detection

Examples: Faiss, Milvus, Pinecone
Remember:
👉 **Vector DB = Similarity Search**

15. Embedded Database

Idea: Database runs **inside the application**, instead of as a separate database server.
Application
↓
Embedded Database

- Fast.
- Small footprint.
- Easy deployment.
- Useful when a separate DB server isn't necessary.

Use cases:

- Desktop applications
- Games
- Local application storage

Examples: SQLite, RocksDB, Berkeley DB
Remember:
👉 **Embedded DB = Database inside the application**